



JABchem



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Past Papers Nat 5 Chemistry

2026 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/125)	%	
A	86+	68.8%	%
B	73+	58.4%	%
C	61+	48.8%	%
D	48+	38.4%	%
No award	<48	<38.4%	%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	15.7 /25	45.2 /75	18.3 /25

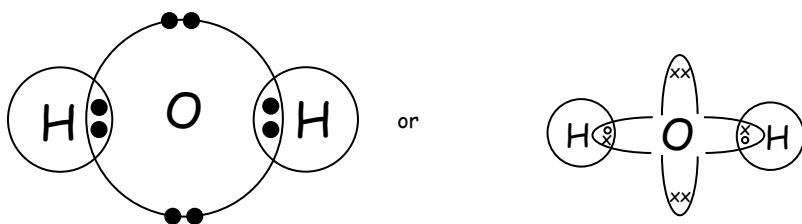
2026 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning								
1	B	74	☒ A Catalysts do not change the mass of product formed ☒ B Catalysts increase reaction rate and <u>reduce</u> the time taken for reaction to finish ☒ C Catalysts are recovered unchanged at the end of a reaction ☒ D Catalysts increase reaction rates								
2	D	58	<table><tr><td>Ammonia NH₃</td><td>phosphorus fluoride PF₃</td></tr><tr><td></td><td></td></tr><tr><td>trigonal pyramidal</td><td>trigonal pyramidal</td></tr></table>	Ammonia NH ₃	phosphorus fluoride PF ₃			trigonal pyramidal	trigonal pyramidal		
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3	C	44	☒ A Zinc powder would give a steeper initial rate than the original zinc lumps ☒ B Mass of zinc reduced to 1g and would result in the final volume being halved. ☒ C Steeper initial rate and final volume half of original. ☒ D Zinc powder would give a steeper initial rate than the original zinc lumps								
4	B	12	☒ A number of electrons in Ne = 10 electrons (1 mol of Ne contains 10mol of electrons) ☒ B number of electrons in F ₂ = 18 electrons (1 mol of F ₂ contains 18mol of electrons) ☒ C number of electrons in O ₂ = 16 electrons (1 mol of O ₂ contains 16mol of electrons) ☒ D number of electrons in N ₂ = 14 electrons (1 mol of N ₂ contains 14mol of electrons)								
5	A	72	Charge = number of protons - number of electrons = 20 - 18 = +2								
6	D	51	☒ A mass number is unchanged ☒ B atomic number (no of protons) does not change when electrons removed to make Na ⁺ ion ☒ C nucleus is unchanged when electrons removed to make Na ⁺ ion ☒ D number of occupied electron shells decreases as Na (2,8,1) becomes Na ⁺ (2,8)								
7	A	70	☒ A Substance A is covalent network as it has high mpt & does not conduct as solid or liquid ☒ B Substance B is ionic as it does not conduct as a solid but does conduct as a liquid ☒ C Substance C is metallic as it conducts as a solid and a liquid ☒ D Substance D is covalent molecular as it has low mpt & does not conduct as solid or liquid								
8	D	59	<table><tr><th>Term</th><th>Meaning</th></tr><tr><td>Solute</td><td>The <i>solid</i> dissolved in the solvent</td></tr><tr><td>Solvent</td><td>The <i>liquid</i> that does the dissolving</td></tr><tr><td>Solution</td><td>The <i>mixture</i> formed when a solute dissolves in a solvent</td></tr></table>	Term	Meaning	Solute	The <i>solid</i> dissolved in the solvent	Solvent	The <i>liquid</i> that does the dissolving	Solution	The <i>mixture</i> formed when a solute dissolves in a solvent
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9	C	68	<table><tr><td>Lithium Li is a metal with r.a.m. of 7 Aluminium Al is a metal with r.a.m. of 27 Total mass of metal in LiAlH₄ = 7+27 = 34</td><td>gfm LiAlH₄ = (1x7) + (1x27) + (4x1) = 7+27+4 = 38</td></tr><tr><td colspan="2">% Mass = $\frac{\text{mass of metal}}{\text{gfm}} \times 100 = \frac{34}{38} \times 100 = 89.5\%$</td></tr></table>	Lithium Li is a metal with r.a.m. of 7 Aluminium Al is a metal with r.a.m. of 27 Total mass of metal in LiAlH ₄ = 7+27 = 34	gfm LiAlH ₄ = (1x7) + (1x27) + (4x1) = 7+27+4 = 38	% Mass = $\frac{\text{mass of metal}}{\text{gfm}} \times 100 = \frac{34}{38} \times 100 = 89.5\%$					
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% Mass = $\frac{\text{mass of metal}}{\text{gfm}} \times 100 = \frac{34}{38} \times 100 = 89.5\%$											
10	A	52	☒ A SO ₂ is a non-metal oxide compound and would dissolve in water to form an acidic solution ☒ B sodium oxide is soluble in water and forms an alkaline pH as sodium oxide is a metal oxide ☒ C Tin(IV) oxide SnO ₂ is insoluble in water and does not change the pH of water ☒ D Silver(I) oxide Ag ₂ O is insoluble in water and does not change the pH of water								
11	B	51	☒ A compound is a pent-2-ene as C=C double bond is between C ₂ and C ₃ number from right ☒ B five carbon mainchain with C=C double bond between C ₂ and C ₃ & -CH ₃ methyl group on C ₄ ☒ C methyl group on C ₄ as numbering of chain decided by numbering C=C double bond from right ☒ D compound is a pent-2-ene as C=C double bond is between C ₂ and C ₃ number from right								

12	D	85	Formula of structure is $C_{10}H_{20}O$ ☒ A $CH_3CH_2CH(OH)CHCHC(CH_3)_3$ has formula $C_9H_{18}O$ ∴ too few carbons to be same structure ☒ B $CH_3CH(CH_3)CH(OH)CH_2CH_2C(CH_3)_3$ has formula $C_{10}H_{22}O$ ∴ too many hydrogens to be same ☒ C $CH_3CH_2CH(OH)CH_2CH_2C(CH_3)_3$ has formula $C_9H_{20}O$ ∴ too few carbons to be same structure ☒ D $CH_3CH(CH_3)CH(OH)CHCC(CH_3)_3$ has formula $C_{10}H_{20}O$ ∴ must be same structure																														
13	B	75	<table><tr><th>Formula</th><th>Mass</th></tr><tr><td>Cyclopropane C_3H_6</td><td>has formula mass = 42</td></tr><tr><td>Cyclobutane C_4H_8</td><td>has formula mass = 56</td></tr><tr><td colspan="2">∴ formula mass INCREASES</td></tr></table>		Formula	Mass	Cyclopropane C_3H_6	has formula mass = 42	Cyclobutane C_4H_8	has formula mass = 56	∴ formula mass INCREASES		<table><tr><th>Strength of Intermolecular Forces</th></tr><tr><td>Strength of Intermolecular forces INCREASES as the carbon chain becomes longer</td></tr></table>	Strength of Intermolecular Forces	Strength of Intermolecular forces INCREASES as the carbon chain becomes longer																		
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14	A	53	☒ A CH_3COOH is ethanoic acid which is diluted and used as vinegar ☒ B CH_3CH_2OH is an alcohol with a -OH hydroxyl group not a -COOH carboxyl group ☒ C CH_3CH_2COOH is propanoic acid and not used in vinegar ☒ D $CH_3CH_2CH_2OH$ is an alcohol with a -OH hydroxyl group not a -COOH carboxyl group																														
15	C	82	<table><tr><th>Ionic bonding</th><th>Covalent Bonding</th><th>Metallic Bonding</th></tr><tr><td>electrostatic force of attraction between oppositely charged ions is called ionic bonding</td><td>electrostatic force of attraction between two nuclei and shared pair of electrons is covalent bonding</td><td>electrostatic force of attraction between positive ions and delocalised electrons</td></tr></table>	Ionic bonding	Covalent Bonding	Metallic Bonding	electrostatic force of attraction between oppositely charged ions is called ionic bonding	electrostatic force of attraction between two nuclei and shared pair of electrons is covalent bonding	electrostatic force of attraction between positive ions and delocalised electrons																								
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16	A	29	☒ A magnesium + hydrochloric acid → magnesium chloride + hydrogen ☒ B magnesium oxide + hydrochloric acid → magnesium chloride + water ☒ C magnesium hydroxide + hydrochloric acid → magnesium chloride + water ☒ D magnesium chloride does not react with acids																														
17	D	83	☒ A sodium is made from electrolysis of a molten sodium compound ☒ B calcium is made from electrolysis of a molten calcium compound ☒ C magnesium is made from electrolysis of a molten magnesium compound ☒ D iron is made by heating iron with carbon/carbon monoxide in a blast furnace																														
18	C	76	☒ A Tin melts at $232^{\circ}C$ and would remain a solid in boiling water at $100^{\circ}C$ ☒ B Indium melts at $157^{\circ}C$ and would remain a solid in boiling water at $100^{\circ}C$ ☒ C Gallium melts at $30^{\circ}C$ and would melt into a liquid in boiling water at $100^{\circ}C$ ☒ D Germanium melts at $938^{\circ}C$ and would remain a solid in boiling water at $100^{\circ}C$																														
19	B	88	<u>Haber Process</u> nitrogen + hydrogen $\xrightarrow{\text{iron catalyst}}$ ammonia <u>Ostwald Process</u> ammonia + Oxygen $\xrightarrow{\text{platinum catalyst}}$ nitrogen monoxide + water ↓ water nitric acid																														
20	C	96	☒ A Potassium is a metal on the left hand side of the Periodic table ☒ B Potassium is a metal found in group 1 of the Periodic Table ☒ C Potassium, phosphorus and nitrogen as all elements essential for healthy plant growth ☒ D Potassium has a symbol K, derived from its original Latin name Kalium																														
21	B	73	<table><tr><th>Area</th><th>Alpha radiation</th><th>Beta Radiation</th><th>Gamma Radiation</th></tr><tr><td>Particle</td><td>helium nucleus ${}^4_2\text{He}$</td><td>Electron ${}^0_{-1}e$</td><td>Electromagnetic radiation</td></tr><tr><td>Mass</td><td>4</td><td>0</td><td>No mass</td></tr><tr><td>Charge</td><td>2+</td><td>1-</td><td>No charge</td></tr><tr><td>Deflection</td><td>Towards negative</td><td>Towards positive</td><td>No deflection</td></tr><tr><td>Stopped By</td><td>Paper</td><td>1cm aluminium</td><td>thick lead</td></tr><tr><td>Use</td><td>Smoke Detector</td><td>Paper thickness in Paper Mill</td><td>Radiotherapy cancer treatment</td></tr></table>	Area	Alpha radiation	Beta Radiation	Gamma Radiation	Particle	helium nucleus ${}^4_2\text{He}$	Electron ${}^0_{-1}e$	Electromagnetic radiation	Mass	4	0	No mass	Charge	2+	1-	No charge	Deflection	Towards negative	Towards positive	No deflection	Stopped By	Paper	1cm aluminium	thick lead	Use	Smoke Detector	Paper thickness in Paper Mill	Radiotherapy cancer treatment		
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22	C	32	☒ A alpha radiation would not travel through paper to detector ☒ B a short half-life would stop the paper thickness measurement as less radiation emitted ☒ C beta radiation would pass through paper and a long half-life would suit long term use ☒ D alpha radiation would not travel through paper to detector
23	A	58	The half-life is the time taken for half the nuclei in a particular isotope to decay. (Answer C assumes that the products formed will not contribute to the remaining mass)
24	A	56	☒ A a more dense brown gas would collect in the measuring cylinder and able to be measured ☒ B a colourless gas would not be able to be viewed so volume could not be measured ☒ C a colourless gas would not be able to be viewed so volume could not be measured ☒ D if the gas is less dense than air then the gas would escape out of the measuring cylinder
25	D	72	☒ A complex X is a negatively charged complex and would take the name cuprate ☒ B Complex Y is a positively charged complex and would take the name iron ☒ C complex X is a negatively charged complex and would take the name cuprate ☒ D Complex X takes the name cuprate and complex Y takes the name iron .

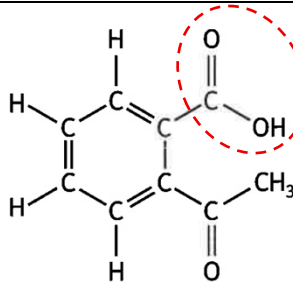
2026 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning								
1a	One answer from:	Same atomic number number of protons Different mass number number of neutrons								
1b(i)	${}^3_1\text{H}$	Mass number (at top) = number of protons + number of neutrons = 1+2 = 3 Atomic number (at bottom) = number of protons = 1 Symbol of element Hydrogen = H								
1b(ii)	Average (masses)	The r.a.m. is the average mass of the naturally-occurring isotopes. As the proportions of each isotope mass is not usually simple proportions like 33%, 50% or 66% then the r.a.m. is unlikely to be a whole number.								
1b(iii)A	20	mass of H₂O water = 2 x mass of deuterium + 1 x mass of oxygen = 2 x 2 + 1 x 16 = 4 + 16 = 20								
1b(iii)B	Diagram showing:									
2a(i)	C ₂ H ₂	General Formula of alkynes = C_nH_{2n-2} Prefix eth- means two carbons If n=2 then 2n-2 = 2 then general formula C_nH_{2n-2} give formula of ethyne = C₂H₂								
2a(ii)	One answer from:	alkene <u>and</u> cycloalkane or alkyne <u>and</u> cycloalkene								
2a(iii)	Cycloalkene	The formula C ₆ H ₁₀ fits the general formula C _n H _{2n-2} and must be either from the alkyne family or the cycloalkene family. As the question states the structure has a ring structure then C ₆ H ₁₀ must belong to homologous series called the cycloalkene.								
2b	One answer from:	Does not contain carbon to carbon double bond. It is saturated. There are only carbon to carbon single bonds.								
2c(i)	3	Formula of zingiberene = C ₁₅ H ₂₄ Three C ₅ H ₈ units join together to make C ₁₅ H ₂₄								
2c(ii)	210	Each H ₂ molecule adds across each of C=C double bonds in zingiberene and the product of addition reaction is the formula of zingiberene with six additional hydrogen atoms. $\begin{array}{ccccc} \text{C}_{15}\text{H}_{24} & + & 3\text{H}_2 & \longrightarrow & \text{C}_{15}\text{H}_{30} \\ 1\text{mol} & & 3\text{mol} & & 1\text{mol} \end{array}$ gfm C₁₅H₃₀ = (15x12) + (30x1) = 180 + 30 = 210g no. of mol C₁₅H₃₀ = 1 mol mass = no. of mol x gfm = 1 x 210 = 210g								
3a	calcium carbonate	<table><tr><th>Ending</th><th>Meaning</th></tr><tr><td>-ide</td><td>2 named elements in compound</td></tr><tr><td>-ate</td><td>2 named elements in compound plus oxygen</td></tr><tr><td>-ite</td><td>2 named elements in compound plus oxygen</td></tr></table> metals always come first in a chemical name.	Ending	Meaning	-ide	2 named elements in compound	-ate	2 named elements in compound plus oxygen	-ite	2 named elements in compound plus oxygen
Ending	Meaning									
-ide	2 named elements in compound									
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3b	$\begin{array}{c} \text{CaCO}_3 + 2\text{HCl} \\ \downarrow \\ \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2 \end{array}$	$\text{CaCO}_3 + 2\text{HCl} \longrightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$								
3c(i)	One answer from:	Stops bubbling Stop fizzing No more gas produced No more reacts No more solid remains								

3c(ii)	2.2	rate = $\frac{\Delta \text{quantity}}{\Delta \text{time}} = \frac{66 - 0}{30 - 0} = 2.2 \text{ cm}^3 \text{ s}^{-1}$																			
3c(iii)	One answer from:	Mass	pH	Concentration	Conductivity																
3c(iv)	One answer from:	Reactant(s) being used up.	Concentration (of acid) decreases	surface area size mass of CaCO_3 mass of limestone } decreases																	
4a	2 different monomers	Problem Solving: Extraction of information from a passage																			
4b	$\begin{array}{c} \text{CH}_3 \\ \\ \text{C}=\text{O} \\ \\ \text{H} \quad \text{O} \\ \quad \\ \text{C}=\text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	Monomers in addition polymerisation must have a $\text{C}=\text{C}$ double bond to be able to polymerise into a polymer.																			
4c	table showing: 1 mark for heading + units 1 mark for correct values	<table><tr><th colspan="2">Percentage of Vinyl Acetate</th><th colspan="2">Elasticity (GPa)</th></tr><tr><td colspan="2">12</td><td colspan="2">91</td></tr><tr><td colspan="2">18</td><td colspan="2">33</td></tr><tr><td colspan="2">28</td><td colspan="2">12</td></tr></table>				Percentage of Vinyl Acetate		Elasticity (GPa)		12		91		18		33		28		12	
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4d	vinyl chloride	monomer	ethene	propene	chloroethene																
		polymer	poly(ethene)	poly(propene)	poly(chloroethene)																
5a	9.61	When mass of $\text{CaCl}_2 = 15\text{g}$, final temperature = $43^\circ\text{C} \therefore \Delta T = 43 - 20 = 23^\circ\text{C}$ heat energy = specific heat capacity $\times$ mass $\times$ change in Temperature $E_h = C \times m \times \Delta T$ $E_h = 4.18 \times 0.1 \times 23$ $E_h = 9.61 \text{ kJ}$																			
5b	Reduce heat loss to surroundings	A polystyrene cup is insulated to reduce heat loss to the surroundings and keep the contents of the cup warmer for longer.																			
5c	Graph showing:	1 mark:	The axes of the graph have suitable scales																		
		1 mark:	All data points plotted accurately and line of best fit added.																		
5d	64 (accept 63-65)																				
6	Open Question:	3 mark answer	2 mark answer	1 mark answer																	
	<table><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>marks</td><td>mark</td><td>marks</td><td>marks</td></tr><tr><td>12%</td><td>44%</td><td>31%</td><td>12%</td></tr></table>	0	1	2	3	marks	mark	marks	marks	12%	44%	31%	12%	Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.					
0	1	2	3																		
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7a(i)	relights a glowing splint	<table><tr><td>Gas</td><td>Hydrogen</td><td>Oxygen</td><td>Carbon Dioxide</td></tr><tr><td>Test</td><td>Burns with a pop</td><td>Relights a glowing splint</td><td>Turns limewater cloudy</td></tr></table>	Gas	Hydrogen	Oxygen	Carbon Dioxide	Test	Burns with a pop	Relights a glowing splint	Turns limewater cloudy							
Gas	Hydrogen	Oxygen	Carbon Dioxide														
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7a(ii)	carbon dioxide and water	Alkanes are a homologous series of hydrocarbons with general formula C_nH_{2n+2} . When hydrocarbons are burned in a plentiful supply of air, the products of complete combustion are carbon dioxide and water.															
7b(i)	As number of branches increases, octane number increases	<table><tr><td>Fuel</td><td>Octane</td><td>2-methylheptane</td><td>2,4-dimethylhexane</td><td>2,2,4-trimethylpentane</td></tr><tr><td>No of branches</td><td>0 methyl groups</td><td>1 methyl group</td><td>2 methyl groups</td><td>3 methyl groups</td></tr><tr><td>Octane Number</td><td>-19</td><td>22</td><td>64</td><td>100</td></tr></table>	Fuel	Octane	2-methylheptane	2,4-dimethylhexane	2,2,4-trimethylpentane	No of branches	0 methyl groups	1 methyl group	2 methyl groups	3 methyl groups	Octane Number	-19	22	64	100
Fuel	Octane	2-methylheptane	2,4-dimethylhexane	2,2,4-trimethylpentane													
No of branches	0 methyl groups	1 methyl group	2 methyl groups	3 methyl groups													
Octane Number	-19	22	64	100													
7b(ii)	2,2,4-trimethylpentane	The higher the octane number the more efficient the fuel.															
7c(i)	Addition	Addition reactions involve a molecule being added across a $C=C$ double bond.															
7c(ii)		Analyse template reaction in question: Apply same changes for new reaction															
8a	uranium & thorium	Problem Solving: Extraction of information from a passage															
8b	UO_2	<table><tr><td rowspan="2">From Passage: Pitchblende is Uranium(IV) oxide</td><td>Write down Symbols and valency below</td><td>Work out chemical formula (Cancel down if necessary)</td></tr><tr><td></td><td>U_2O_4 cancels down to UO_2</td></tr></table>	From Passage: Pitchblende is Uranium(IV) oxide	Write down Symbols and valency below	Work out chemical formula (Cancel down if necessary)		U_2O_4 cancels down to UO_2										
From Passage: Pitchblende is Uranium(IV) oxide	Write down Symbols and valency below	Work out chemical formula (Cancel down if necessary)															
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8c	276 days	<table><tr><td>No. of half-lives</td><td>Percentage of Original Radioactivity</td></tr><tr><td>0</td><td>100%</td></tr><tr><td>1</td><td>50%</td></tr><tr><td>2</td><td>25%</td></tr></table> From passage: 1 half-life for polonium = 138 days $\therefore$ 2 half-lives = 2×138 days = 276 days	No. of half-lives	Percentage of Original Radioactivity	0	100%	1	50%	2	25%							
No. of half-lives	Percentage of Original Radioactivity																
0	100%																
1	50%																
2	25%																
8d	Atomic number No. of protons } decreased	Uranium has an atomic number of 92, thorium has an atomic number of 90 and radium has an atomic number of 88. As alpha decay reduces atomic number by 2, uranium will alpha decay into a Thorium and a thorium will alpha decay into Radium.															
9a(i)	One answer from:	<table><tr><td>Aluminium's value should be larger than that of zinc</td><td>Aluminium is higher than zinc in the ECS but has a lower voltage.</td><td>The order should be magnesium, aluminium, zinc, iron (or the reverse)</td></tr></table>	Aluminium's value should be larger than that of zinc	Aluminium is higher than zinc in the ECS but has a lower voltage.	The order should be magnesium, aluminium, zinc, iron (or the reverse)												
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9a(ii)	ions free to move	The electrolyte in an electrochemical cell completes the circuit as ions move through the electrolyte to balance the movement of electrons through the wires.															

9b(i)	arrows on wires from Zinc to Copper	Electrons always from a metal higher up the electrochemical series to the metal lower down the electrochemical series. Zinc is higher up metal in ECS giving oxidation reaction: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ Copper is lower down metal in ECS giving reduction reaction: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$				
9b(ii)	to complete the circuit or to allow ions to flow	As negatively-charged ions move from the zinc electrode through the wires to the copper electrode, the buildup of negative charge is balanced out by the movement of ions through the ion bridge to keep both sides electrically neutral.				
9c(i)	1.10	$\begin{aligned} E^0_{\text{cell}} &= E^0_{(\text{reduction})} - E^0_{(\text{oxidation})} \\ &= E^0 \text{ from Copper} - E^0 \text{ from Zinc} \\ &= 0.34 - (-0.76) \\ &= 1.10\text{V} \end{aligned}$				
9c(ii)	$\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$	Oxidation $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$ Reduction $\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$ $\text{Zn} \longrightarrow \text{Zn}^{2+} + \cancel{2\text{e}^-}$ $\text{Cu}^{2+} + \cancel{2\text{e}^-} \longrightarrow \text{Cu}$ Redox $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$				
9c(iii)	39.9	no. of mol = volume $\times$ concentration = $0.25\text{litres} \times 1.0\text{mol l}^{-1} = 0.25\text{mol}$ gfm $\text{CuSO}_4 = (1 \times 63.5) + (1 \times 32) + (4 \times 16) = 63.5 + 32 + 64 = 159.5\text{g}$ mass = no. of mol $\times$ gfm = $0.25 \times 159.5 = 39.9\text{g}$				
10a(i)	Answer containing:	Family of compounds with same/similar chemical properties <u>and</u> a general formula				
10a(ii)	hydroxyl (group)	<table><tr><td>$-\text{O}-\text{H}$</td><td>$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>	$-\text{O}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$	hydroxyl group	carboxyl group
$-\text{O}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$					
hydroxyl group	carboxyl group					
10b	11.2	$700\text{cm}^3 \text{ whisky} \longleftrightarrow 224\text{g ethanol}$ $35\text{cm}^3 \text{ whisky} \longleftrightarrow 224\text{g ethanol} \times \frac{35}{700}$ $= 11.2\text{g ethanol}$				
10c(i)	formulae written with equilibrium arrow	$\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \rightleftharpoons \text{C}_3\text{H}_6\text{O}_2 + \text{H}_2\text{O}$ methanol ethanoic acid methyl ethanoate water				
10c(ii)	use hot water bath or electric hotplate	Both reactants are flammable and naked flames should not be used near the reaction as vapours of the reactants could ignite. A hot water bath, boiling water from a kettle or an electric hot plate could be used.				
10c(iii)	<u>Alcohol</u> propan-1-ol <u>Carboxylic Acid</u> butanoic Acid	alcohol side $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C}- & \text{C}- \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$ made from 3 carbon alcohol carboxylic acid side $\begin{array}{c} & \text{O} & & \text{H} & \text{H} & \text{H} \\ & & & & & \\ & \text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{H} \\ & & & & \\ & & \text{H} & \text{H} & \text{H} \end{array}$ made from 4 carbon carboxylic acid				

10c(iv)		Acids have pH less than 7. Carboxylic Acids are acids with a pH less than 7. The carboxyl -COOH group is the functional group in carboxylic acids.										
11a(i)	exothermic	<table border="1"><thead><tr><th>Term</th><th>Meaning</th></tr></thead><tbody><tr><td>exothermic</td><td>Reaction or process which releases heat energy</td></tr><tr><td>endothermic</td><td>Reaction or process which takes in heat energy</td></tr></tbody></table>			Term	Meaning	exothermic	Reaction or process which releases heat energy	endothermic	Reaction or process which takes in heat energy		
Term	Meaning											
exothermic	Reaction or process which releases heat energy											
endothermic	Reaction or process which takes in heat energy											
11a(ii)	lilac	The only metal in the reactants is potassium which gives a lilac flame colour.										
11b	H ⁺ ions present	All acids contain H ⁺ hydrogen ions. If H ⁺ ions are reactants in an chemical reaction then adding acid to the reaction mixture will allow the reaction to proceed if all reactants are present.										
11c(i)	21.4	Ignore 1 st Titre (rough titre) as it is not concordant (not within 0.2cm ³) average titre = $\frac{21.3 + 21.5}{2} = \frac{42.8}{2} = 21.4\text{cm}^3$										
11c(ii)A	0.02675	no. of moles = volume x concentration = 0.0214 litres x 0.005 mol l ⁻¹ = 0.000107 mol $5\text{Fe}^{2+} + \text{MnO}_4^- + 8\text{H}^+ \longrightarrow 5\text{Fe}^{3+} + \text{Mn}^{2+} + 4\text{H}_2\text{O}$ 5mol 0.000535mol 1mol 0.000107mol concentration = $\frac{\text{no. of moles}}{\text{volume}} = \frac{0.000535 \text{ mol}}{0.020 \text{ litres}} = 0.02675 \text{ mol l}^{-1}$										
11c(ii)B	it changes colour or there is colour change or self-indicating	$5\text{Fe}^{2+} + \text{MnO}_4^- + 8\text{H}^+ \longrightarrow 5\text{Fe}^{3+} + \text{Mn}^{2+} + 4\text{H}_2\text{O}$ purple colourless The purple colour of permanganate MnO ₄ ⁻ ions is very different to the colour of the Manganese Mn ²⁺ ions formed during the reaction. No indicator is required in this titration if the reactants are naturally a different colour to the products.										
12	Open Question:		3 mark answer	2 mark answer								
			1 mark answer									
	<table border="1"><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>6%</td><td>20%</td><td>50%</td><td>6%</td></tr></table>	0 marks	1 mark	2 marks	3 marks	6%	20%	50%	6%	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
0 marks	1 mark	2 marks	3 marks									
6%	20%	50%	6%									